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Assignment - Number system

Q1. Complete the table with equivalents in othere number system for $143.75_{(10)}$, $11011.101_{(2)}$, $453_{(8)}$, $1F.C_{(16)}$

	Decimal	Binary	Octal	Hexa
1	143.75	10001111.11	217.6	8F.C
2	27.625	11011.101	33.5	1B.A
3	299	100101011	453	12B
4	31.75	11111.11	37.6	1F.C

Q2. Find equivalents:

a) $324_5 = ?_{10}$

$= 3(5^2) + 2(5^1) + 4(5^0)$

$= 89_{10}$

b) $32.23_4 = ?_{10}$

$= 3(4^1) + 2(4^0) + 2\left(\frac{1}{4}\right) + 3\left(\frac{1}{4^2}\right)$

$= 14.6875_{10}$

c) $231_{10} = ?_7$

7	231
7	33 → 0
	4 → 5

$231_{10} = 450_7$

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d) $189_{10} = ?_6$

$$\begin{array}{r|l} 6 & 189 \\ 6 & 31 \rightarrow 3 \\ & 5 \rightarrow 1 \end{array}$$

$189_{10} = 513_6$

Q3. Find the radix r:

a) $(312/20)_r = 13.1_r$

$$\frac{3r^2 + 1r + 2}{2r} = 1r + 3 + \frac{1}{r}$$

$$3r^2 + 1r + \cancel{2} = 2r^2 + 6r + \cancel{2}$$

$$r^2 = 5r$$

$$r = 5$$

b) $264_r + 136_r = 433_r$

$$2r^2 + 6r + 4 + 1r^2 + 3r + 6 = 4r^2 + 3r + 3$$

$$3r^2 + 9r + 10 = 4r^2 + 3r + 3$$

$$6r + 7 = r^2$$

$$r^2 - 6r - 7 = 0$$

$$r = 7$$

or

we can observe that $4_r + 6_r = 13_r$

$$4 + 6 = r + 3$$

$$\boxed{r = 7}$$

c) $221505_8 = 12345_r$

$221505_8 = 2(8^5) + 2(8^4) + 1(8^3) + 5(8^2) + 0 + 5$

$= 74565$

$74565 = 1(r^4) + 2(r^3) + 3(r^2) + 4(r) + 5$

$16^4 = 65,536$

so try $r = 16$

$= 65536 + 8192 + 768 + 64 + 5$

$= 74565$

$\therefore r = 16$

Q4. Addition:

a) $1101_2 + 11011111_2$

$$
\begin{array}{r}
1101 \\
11011111 \\
\hline
11101100
\end{array}
$$

$= 11101100_2$

b)
$$
\begin{array}{r}
77652_8 \\
+\ 76543_8 \\
\hline
176415_8
\end{array}
$$

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c) $\begin{array}{r} 578A_{16} \\ +\ ABDE_{16} \\ \hline 10368_{16} \end{array}$

d) $234_6 + 315_6$

$\begin{array}{r} 234_6 \\ +\ 315_6 \\ \hline 553_6 \end{array}$

Q5. Binary → Gray Code:

a) 10101_2
⇒ 11111_{gray}

b) 01110_2
⇒ 01001_{gray}

Q6. Gray Code → Binary

a) 01101_{gray}
⇒ 01001_2

b) 10101_{gray}
⇒ 11001_2